

Qingdao University of Technology  
Qingdao, Shandong, China

11 August 2026

Editor-in-Chief

*Concurrency and Computation: Practice and Experience*

Dear Editor-in-Chief,

We submit the manuscript entitled “Concurrency-Aware Permission Decisions in Multi-Tenant LLM Agent Runtimes” for consideration as a research article in *Concurrency and Computation: Practice and Experience*.

The manuscript studies a concurrency failure in a multi-tenant agent runtime: tenant work can fill a bounded executor and delay a permission decision until its deadline, while a permissive timeout fallback converts executor unavailability into unsafe admission. We present Neuron, a Java 21 runtime architecture that reserves permission capacity and exports a resource-pressure signal to the escalation policy. The paper’s primary contribution is a systems characterization of executor contention, queueing, deadline semantics, and permission-verdict behavior; the security consequence is the measured failure mode rather than a claim of a new authorization theory.

The evaluation compares shared fail-open, shared fail-closed, isolated, and coupled architectures. It contains five independent JVM runs on two recorded environments, deadline and attacker-pressure surfaces, capacity sweeps, mixed load QoS, fixed API-trace replay, a process-separated Java permission service, and an additional executor-focused Windows replication. Raw JSONL observations, environment records, analysis scripts, generated values, and reproduction instructions are publicly archived at <https://github.com/dadawer-web/ouisani/tree/computing-repro-20260810/computing-repro-20260810>. The process-separated service is explicitly a same-host loopback experiment; we do not claim WAN, multi-node, GPU, or live-production behavior.

We believe the manuscript fits the journal’s emphasis on practical concurrent and distributed systems because it reports an implemented Java runtime, executor-level measurements, architecture baselines, and reproducible engineering trade-offs. The manuscript is original, is not under consideration elsewhere, and has been approved by both authors.

Thank you for your consideration.

Sincerely,

Mingyuan Xie  
Zhengxun Wu  
Qingdao University of Technology  
Corresponding authors  
jhj208436@gmail.com  
2276815088@qq.com